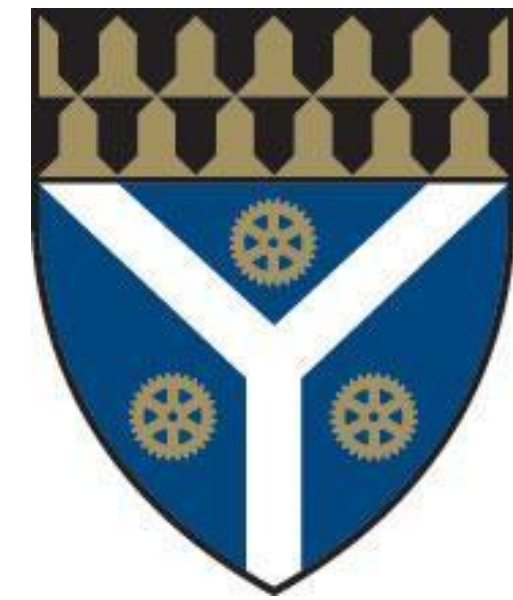




Applying score-following tools to visualize expressive timing in music

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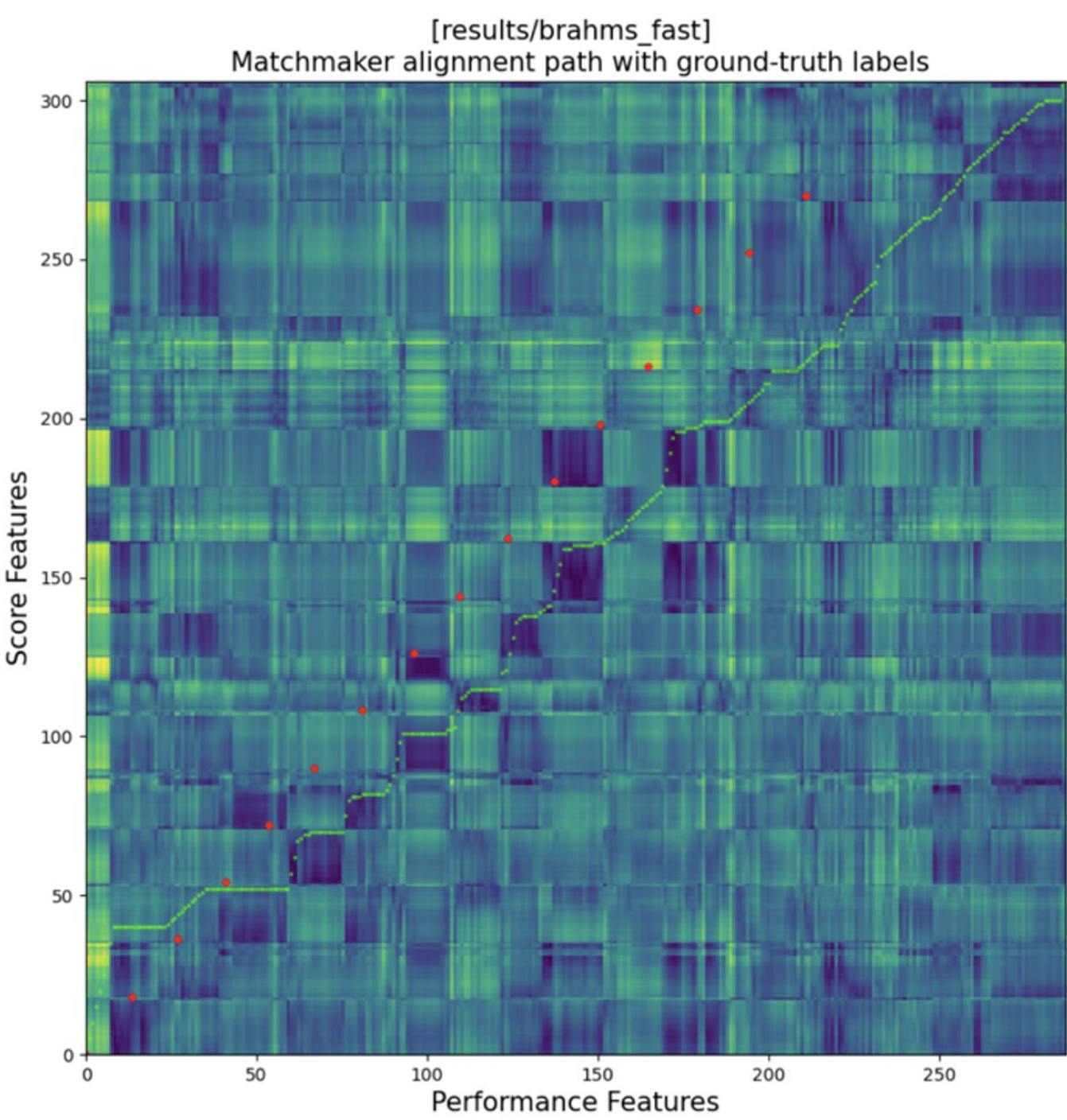
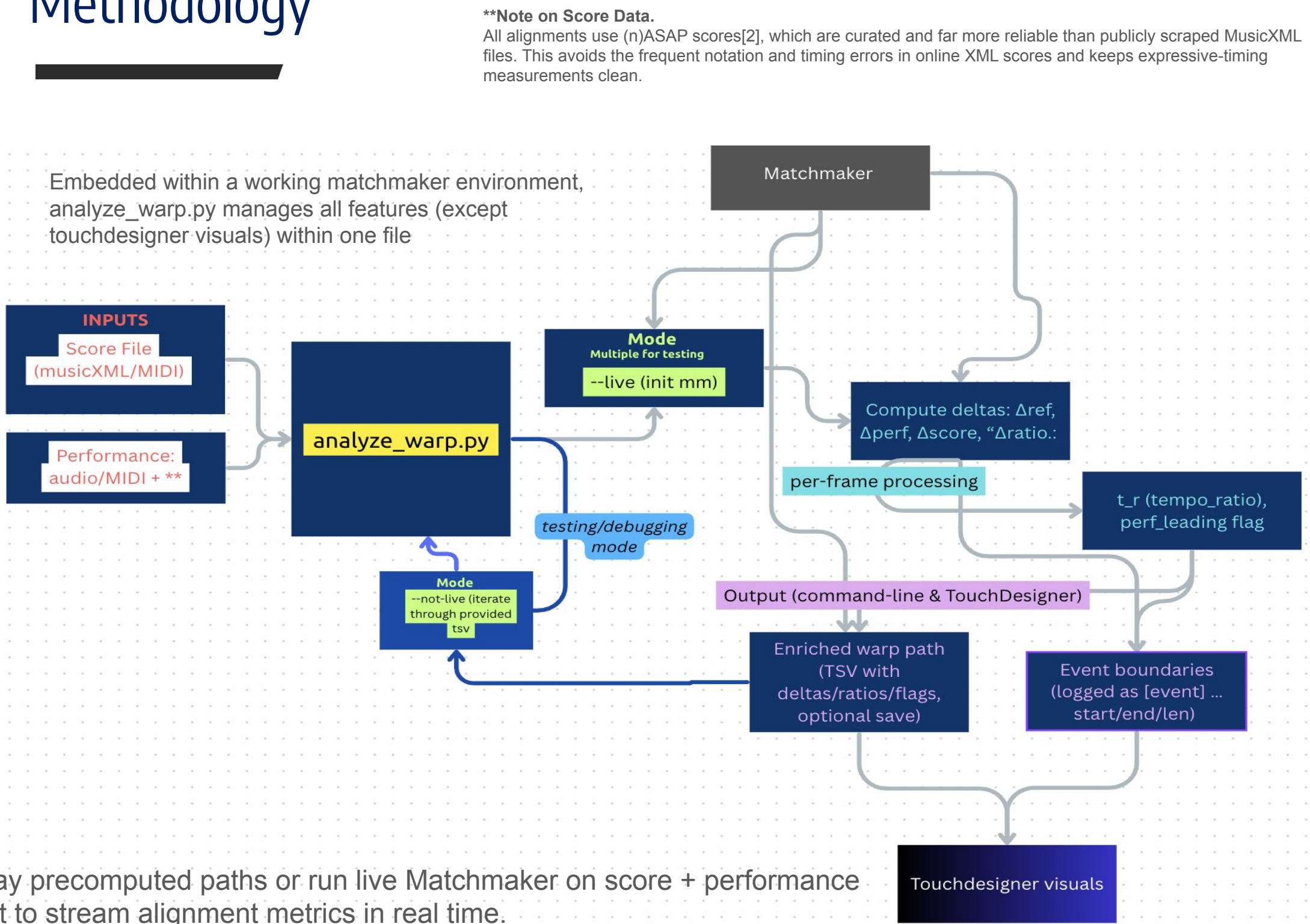


Purpose/Background

Musical expressivity often manifests through timing deviations (speeding up, slowing down) and **knowing when/how** it is done (and whether it is intentional or not) is important.

Score following aligns a live performance to a notated score in real time. Used in computer-assisted performance, interactive media, and analysis of recordings. Expressive variations—rubato, hesitations, accelerations—can cause misalignment or tracking drift. Handling timing deviations requires modeling nonlinear time-warping between score and performance. Recent methods (e.g., **Matchmaker**[3], a python library of alignment algorithms) use dynamic alignment to better tolerate variation—while offering insight on unexpected changes in timing. **This project aims to use the existing features of this already implemented framework as a tool to display these events through text and visuals**

Methodology



Alignment of Brahms I (fast performance): warp path showing score vs. performance frames, with perf-leading segments flagged and score jump regions visible in the path slope.

How can we automate expressive timing event detection?

Start from frame indices: ref_idx (score), $input_idx$ (performance).

Convert to time: $score_time = ref_idx / frame_rate$, $perf_time = input_idx / frame_rate$.

Frame-to-frame deltas: $\Delta ref_idx = \Delta ref_idx$, $\Delta input_idx = \Delta input_idx$; $\Delta score_time = \Delta score_time$, $\Delta perf_time = \Delta perf_time$.

Path slope (gradient): $\Delta ratio = \Delta perf / \Delta ref$ when $\Delta ref \neq 0$; if $\Delta ref = 0$ and $\Delta perf > 0$, use $\Delta perf$ to mark perf advancing while score is clamped.

Tempo ratio: $tempo_ratio = \Delta perf / \Delta score$ when $\Delta score \neq 0$ (otherwise nan); perf_leading flag when $\Delta ref = 0$ and $\Delta perf > 0$. [1]

replay precomputed paths or run live Matchmaker on score + performance input to stream alignment metrics in real time. Enriched each frame with temporal deltas, ratios, and explicit perf-leading/score-jump flags; immediate event logging captures start/end/len for downstream visualization. Outputs stay analysis-ready: saved TSVs with derived fields, evaluation metrics/PNGs (when annotated), and summaries to feed TouchDesigner or other frontends. **Design goal:** make alignment behavior legible—no hidden NaNs, clear markers for fast/slow sections, and zero-cost post-processing without rerunning alignment.

Results/Conclusion

Extracted frame-level timing features (Δref , $\Delta perf$, $\Delta time$, tempo ratios) from Matchmaker's score-performance alignment output.

Implemented rule-based detectors that flag score-jumps, performance-leads, non-advancing score frames, and other alignment irregularities directly from $\Delta ref/\Delta perf$ patterns.

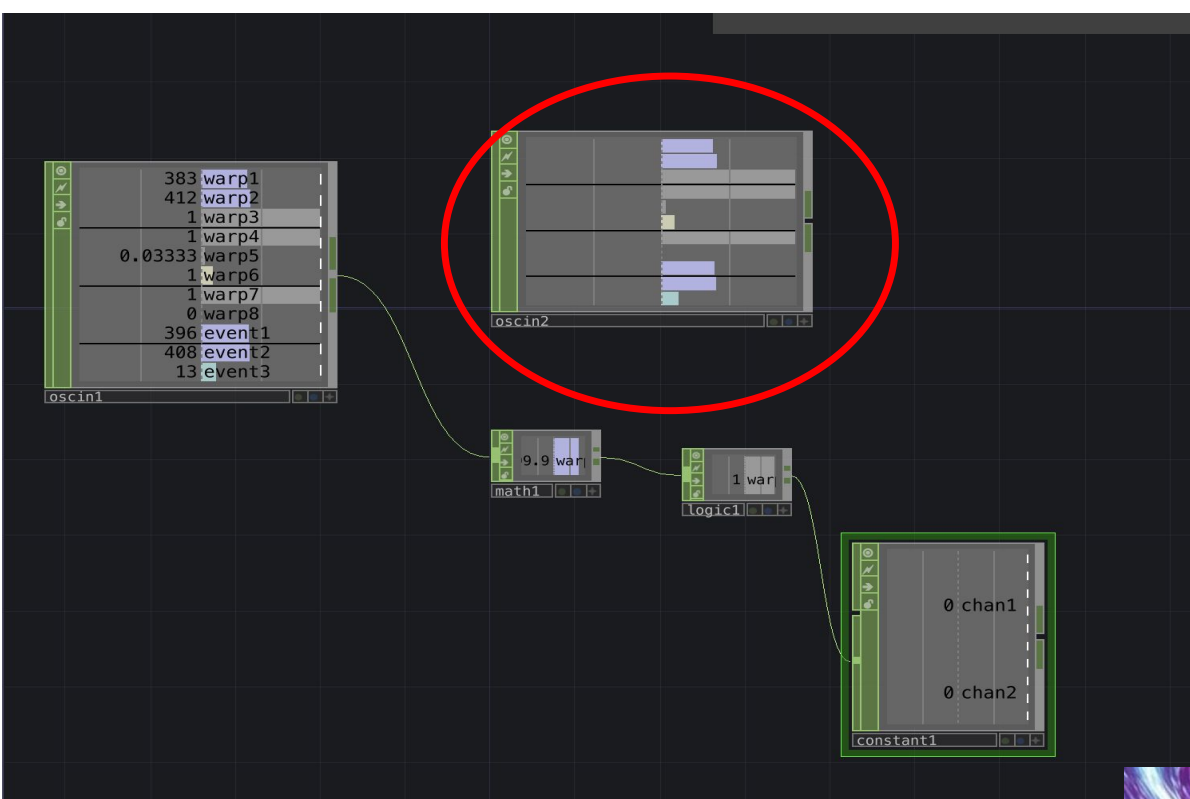
Computed slope trends in tempo-ratio. Grouped consecutive frames into contiguous timing-event segments, producing clearer rubato/flow patterns and reducing frame-level noise.

Generated event-annotated TSVs that summarize timing deviations across an entire performance for quick inspection or downstream analysis.

Built a pipeline to stream timing events as OSC messages into TouchDesigner, enabling real-time audiovisual mappings of expressive timing.

Although earlier planning considered adding hysteresis-based smoothing to stabilize event detection, the final system did not require it: once $\Delta ref/\Delta perf$ patterns and tempo-ratio slopes were grouped into contiguous runs, the events proved naturally robust without additional threshold-memory. This allowed the post-processing pipeline to stay lightweight, transparent, and directly interpretable while still capturing meaningful expressive-timing structure.

Scan for video demos



Current visualization mechanism: OSC CHOP in TouchDesigner feeding into constant CHOPS to show pulsating bars



Future Work: MUSI4996 Continuation

Develop a more user-friendly interface (GUI for score input, playback controls, parameter tuning).

Implement real-time visualization of timing deviations and alignment.

Extend system beyond piano to non-piano instruments.

Begin adapting the pipeline for ensemble score-following, starting with a string quartet.

Scale to larger ensembles (full orchestra) with multiple-part tracking. (If unsuccessful, put on solo piano recital with visualizations by this working)

Aim to produce a live performance using the working application.

Bibliography

[1] Cont, Arshia. "ANT: A Real-Time Framework for Score Following." Proceedings of the Sound and Music Computing Conference, 2008.

[2] Peter, S.D., Cancino-Chacón, C.E., Foscari, F., McLeod, A.P., Henkel, F., Karystinaios, E. and Widmer, G. (2023) 'Automatic Note-Level Score-to-Performance Alignments in the ASAP Dataset', Transactions of the International Society for Music Information Retrieval, 6(1), p. 27–42.

[3] Park, J., Cancino-Chacón, C., Chiruthapudi, S., & Nam, J. (2021). MatchMaker: An open-source library for real-time piano score following and systematic evaluation. Proceedings of the 22nd International Society for Music Information Retrieval Conference (ISMIR 2021).